

Name: _____

Student ID: _____

Signature: _____

UNSW Sydney
TERM 1 2025 MIDTERM EXAMINATION
ELEC3115: Electromagnetic Engineering

1. TIME ALLOWED – 1 hour
2. READING TIME – 10 minutes (includes in 1 hour)
3. THIS EXAMINATION PAPER HAS 4 PAGES
4. TOTAL NUMBER OF QUESTIONS – 3
5. TOTAL MARKS AVAILABLE – 50
6. MARKS AVAILABLE FOR EACH QUESTION ARE SHOWN IN THE EXAMINATION PAPER
7. ALL ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY BE USED ONLY FOR DRAWING, SKETCHING OR GRAPHICAL WORK
8. THIS PAPER MAY **NOT** BE RETAINED BY CANDIDATE
9. CANDIDATES MAY BRING TO THE EXAMINATION an UNSW-approved calculator
10. Any assumptions found to be necessary in answering the questions should be stated explicitly. Must show all your workings to receive any partial marks applicable.

Question 1 [Marks 20] (Topic 1)

- (a) [5 marks] Explain briefly how Gauss's law is related to the first postulate of electrostatics. Under what conditions is Gauss's law instrumental in determining the electric field intensity of a uniform charge distribution?
- (b) [4 marks] Define the polarisation vector. What is its SI unit? Observing the SI unit, what can we deduce about the physical nature of the polarisation vector?
- (c) [6 marks] A parallel-plate capacitor with a 0.5 m^2 area and a plate separation of 30 mm is filled with two dielectric slabs stacked side by side. The relative permittivity of the left-hand side and right-hand side slabs are 6 and 12, respectively. If a potential of 200V is applied to the capacitor, find
- (i) the polarisation vector in each slab.
 - (ii) the total capacitance of the capacitor.
- (d) [5 marks] State how the normal and tangential components of the electric field intensity at the boundary of the two dielectric slabs of part (c) are related.

Question 2 [Marks 20] (Topic 2)

- (a) ~~[7 marks]~~ ^[10 marks] A copper track on a multilayered printed circuit board (PCB) is 0.5 mm wide and separated from the ground plane by 1 mm. The relative dielectric constant of the homogeneous PCB material (FR-4 epoxy-resin glass-fiber) is $\epsilon_r = 2.5$. The arrangement is depicted in Figure Q2.

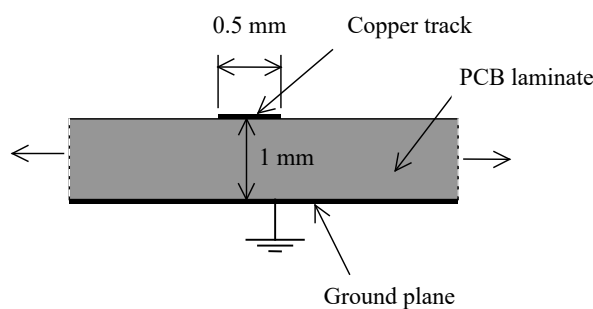


Figure Q2

Using the method of images,

- (i) [5 marks] Calculate the capacitance per meter between the copper track and the ground plane.
- (ii) ~~[2 marks]~~ ^[5 marks] Why can one not use the method of images for this problem if the ground plane has a finite extent?

[10 marks]

- (b) ~~[6 marks]~~ Derive Poisson's equation from the first postulate of the electrostatics. List the various methods for solving Poisson's equation and specify the conditions under which each method is useful.

Question 3 [Marks 10] (Topic 3)

- (a) [6 marks] A toroidal coil is used to store energy. The cross-sectional radius of the circular core of the toroid is 1m, and the mean radius of the torus is 1 km.
- (i) Find the reluctance of the torus core if it is made of magnetic steel with a relative permeability of 5000.
 - (ii) If the number of turns in the coil is 100 and carries a current of 100 A, calculate the flux in the core using the *magnetic circuit method*.
 - (iii) How much magnetic energy can be stored in this toroidal inductor?
- (b) [4 marks] What is a magnetic dipole? using the concept of magnetic dipole, discuss the magnetisation of ferromagnetic materials.

END OF EXAM PAPER

Useful formula and constants:

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C/m}$$

$$\vec{\nabla} \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\vec{\nabla} \times \vec{A} = \vec{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \vec{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \vec{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{Q_{enclosed}}{\epsilon} \quad \text{or} \quad \oint_s \vec{D} \cdot d\vec{s} = Q_{enclosed}$$

$$Q_{enclosed} = \int_v \rho dv \quad \text{or} \quad \int_s \rho_s ds \quad \text{or} \quad \int_l \rho_l dl$$

$$V_{ba} = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E}, \quad \vec{D} = \epsilon \vec{E}, \quad \epsilon = \epsilon_0 \epsilon_r$$

$$\vec{E} = \left| \frac{\rho_l}{2\pi\epsilon r} \right| \vec{a}_r, \quad V/m, \quad V_{21} = \frac{\rho_l}{2\pi\epsilon} \ln \left(\frac{r_1}{r_2} \right)$$

$$|E_{1t}| = |E_{2t}|, \quad |D_{1n}| = |D_{2n}|$$

$$C = \frac{\pi\epsilon}{\ln \frac{d-a}{a}} \quad F/m, \quad C = \frac{2\pi\epsilon}{\ln \left(\frac{b}{a} \right)} \quad F/m$$

$$w_e = \frac{1}{2} \vec{D} \cdot \vec{E} \quad [J/m^3], \quad \vec{F}_e = \pm \vec{\nabla} W_e$$

$$\vec{\nabla}^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \varphi^2} + \frac{\partial V^2}{\partial z^2} = 0$$

$$C = \frac{\pi\epsilon_0}{\ln \left[\left(\frac{d}{2a} \right) + \sqrt{\left(\frac{d}{2a} \right)^2 - 1} \right]} \quad F/m$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\oint_c \vec{B} \cdot d\vec{l} = \mu_0 I_c, \quad \oint_c \vec{H} \cdot d\vec{l} = I_{enclosed}$$

$$\vec{B} = \vec{\nabla} \times \vec{A},$$

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{M}, \quad \vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$$

$$\vec{\nabla}^2 \vec{A} = -\mu_0 \vec{J}$$

$$\Phi = \int_s \vec{B} \cdot d\vec{s}, \quad \Re = \frac{l}{\mu S},$$

$$L = \frac{\lambda = N\phi}{i},$$

$$EMF = -N \frac{d\phi}{dt}, \quad EMF = \oint_c (\vec{u} \times \vec{B}) \cdot d\vec{l}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$V_{rms} = 4.44 f N B_m S_c$$

$$\delta = \sqrt{\frac{2}{\sigma\mu\omega}}, \quad R_{ac} = \frac{l}{\delta\sigma} \left(\frac{l}{\pi(D-\delta)} \right)$$

$$P_h = k_2 B_m^n f, \quad P_e = k_1 B_m^2 f^2$$

$$\vec{J} = \rho \vec{u}, \quad I = \int_s \vec{J} \cdot d\vec{s}$$

$$\vec{J} = \sigma \vec{E}, \quad R = \frac{l}{\sigma S}, \quad P = \int_{v'} (\vec{E} \cdot \vec{J}) dv'$$

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}, \quad \rho = \rho_0 e^{-\left(\frac{\sigma}{\epsilon}\right)t}$$

$$\vec{\nabla} \times \vec{J} = 0$$

$$RC = \frac{\epsilon}{\sigma}$$